PROJECT OVERVIEW

Jejak 42·50

A training-plan and running-journal app that builds its own programme from race dates, reads watch data, and warns before injuries happen, for the road from a half marathon to a full marathon and a 50 km ultra.

Raihan Evanza · Fullstack & Mobile Developer · portfolio-raihan-evanza.vercel.app

ROLE

Personal project · design, frontend, backend, deployment · used daily by me

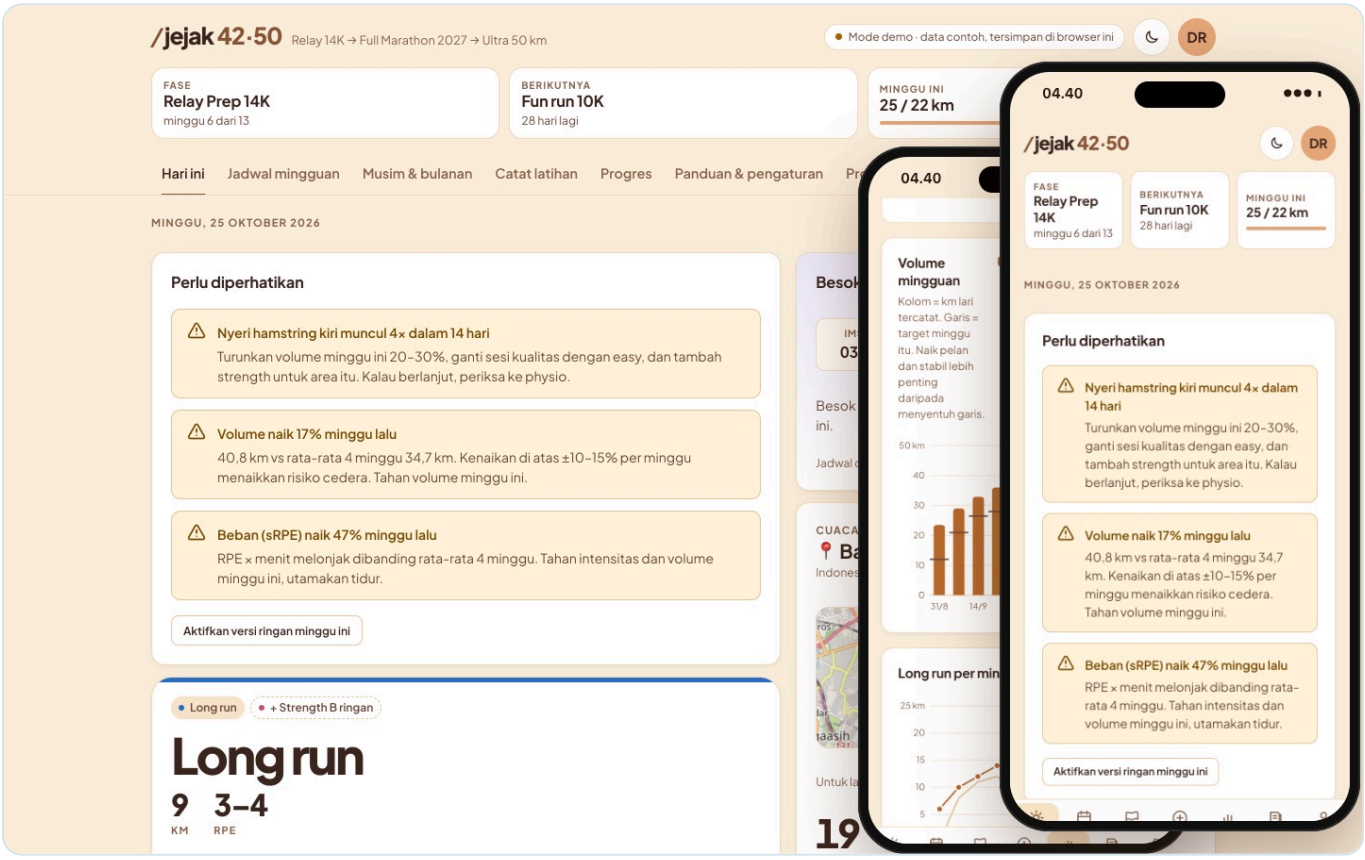
PLATFORMS

PWA (iPhone, Android, desktop), installable on the Home Screen and usable offline

LINKS

Demo (sample data): raihan-evanza-jejak.vercel.app

React 19 · Vite · Rupa Design System · PWA + Service Worker · Vercel Functions & Cron · Supabase Auth & Postgres · Gemini API · intervals.icu API · Web Push · jsPDF · OpenStreetMap · Open-Meteo



OVERVIEW

Typical running apps record what already happened, but they don't build a programme toward several goals at once: a 14 km relay leg, a full marathon, mountain hikes, then a 50 km ultra. A spreadsheet plan also can't tell when sleep is short, pain keeps coming back in the same spot, or volume climbs too fast.

Jejak 42·50 generates a weekly plan from the main races and adjusts it using a morning check-in, logged sessions, and watch data (Huawei via intervals.icu). It warns about training load, predicts race times, builds a race-day plan, and has an AI coach (Gemini). The plan can be changed one day at a time or a whole month at once with the month planner.

It started as a single index.html for one person behind one password. It is now a React 19 frontend on the Rupa design system (latte brand), 12 Vercel Functions, and Supabase: everyone has their own account, row-level security guards

the data, and a superadmin manages accounts without being able to open personal data. I use it every day; the portfolio video and screenshots use demo data.

KEY FEATURES

Plan generated from the main races

Every race and hike is an event; three are marked as main races (relay, full marathon, ultra). Training phases are built backwards from their dates, with quality sessions, a long run, easy runs, and strength A/B each week, including adjustments for Ramadan and fasting days. The ultra phase scales with the distance, from a 50K build up to 180K (long runs of about 40 km, back-to-backs, and night runs).

Month planner

Besides the weekly view, the plan opens as a month: pick a menu (easy, tempo, long run, strength, rest, event) and tap the dates to change. Tapping a date with no menu selected opens that day.

Morning check-in & warnings

Sleep, stress, soreness, and pain (with a body map) are logged each morning. The app flags volume rising more than 15%, sRPE load spikes, and pain recurring in the same area, then offers a lighter version of the rest of the week.

Three ways to log a session

The manual form works out pace and category from HR and pace. Huawei Health screenshots are read by Gemini (or Tesseract OCR in the browser). Watch sessions arrive by themselves through intervals.icu and merge with a similar manual entry instead of duplicating it.

Session detail from the watch

For synced sessions: a route map (OpenStreetMap), pace, HR, cadence, and elevation charts across the run, time in each HR zone, and per-km splits. A Share button renders an image card for WhatsApp or stories.

Progress & race predictions

Charts for weekly volume, long runs, easy-run HR, VO2max, and Fitness · Fatigue · Form; achievements and streaks; Road to race; monthly recaps; sRPE load; and race-time predictions against targets.

Race-day plan

For each race: pace per segment from the target time, a gel and drink schedule by minute, and an hourly fluid target from the sweat-rate calculator in the profile. The page can be printed or saved as PDF.

Accounts & superadmin

A new sign-in page on Supabase Auth. A superadmin adds, edits, resets, and deletes accounts, revokes coach and calendar links, and reads each account's run history (read-only). Check-ins, pain, profile, and plans stay private.

AI coach, reminders & reports

Daily, weekly, and monthly reviews by Gemini plus free-form questions; morning and evening push notifications via Vercel Cron; an .ics calendar feed; a read-only coach link; a PDF report; JSON backup and CSV export; light and dark themes.

ARCHITECTURE & DATA FLOW

PWA (React)
UI and planning logic in the browser; signs in straight to Supabase Auth; write queue while offline

→

Vercel Functions
12 functions: data, doc, account, admin, intervals, import, review, ask, push, calendar, coach, config

→

Supabase Postgres
A docs table: one row per record (account + collection + id + JSON), guarded by RLS

→

Layanan luar
intervals.icu (Huawei watch), Gemini, Web Push, Open-Meteo, OpenStreetMap

- The page loads all of an account's data once through GET /api/data, then sends each change as a single record through POST /api/doc with the Supabase access token. Functions pass that token on to Supabase, so row-level security makes sure each account only reads and writes its own rows.
- The superadmin reads run history through the admin_user_logs SQL function (security definer, role checked in the database), which returns only the running part of each session. Accounts are created by the superadmin; open sign-up is off.
- The service worker caches the app shell, while /api is never cached. Writes made offline wait in a queue and go out once online. The Supabase secret key, Gemini, and VAPID keys live only in Vercel env.

STACK BY LAYER

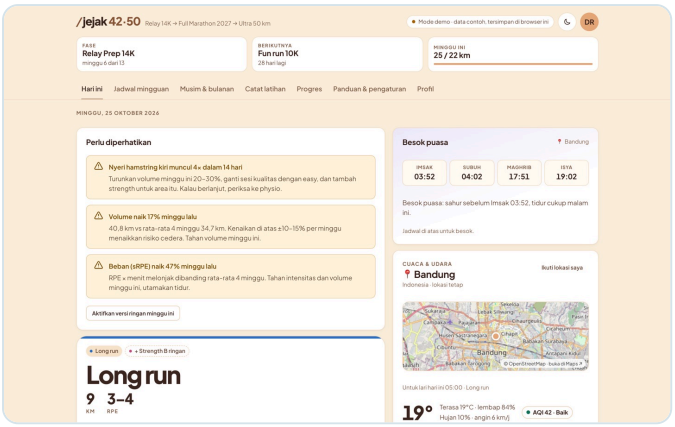
Frontend	React 19 + Vite · Rupa Design System (@rupa-ui/react, @rupa-ui/tokens, brand latte) · Grafik SVG buatan sendiri · Tema terang & gelap
PWA	Service Worker · Web App Manifest · Web Push (VAPID) · Antrean tulis offline
Backend	Vercel Functions (Node.js) · Vercel Cron · web-push
Data & accounts	Supabase Auth · Supabase Postgres (RLS, fungsi SQL) · Tes RLS di Docker · Backup JSON, ekspor CSV
Integrations	intervals.icu API · Gemini API (REST) · Tesseract.js (OCR cadangan) · Open-Meteo · OpenStreetMap
Documents	jsPDF (dimuat saat dibutuhkan) · Feed kalender .ics

TECHNICAL DECISIONS

1. **From one file to React + Rupa**
   The first version was a single framework-free index.html. When accounts, a superadmin, and the month planner arrived, it was split up: the planning logic moved unchanged into src/lib, each tab became a React component, and every color and base component comes from Rupa to stay consistent with my other projects.
2. **Supabase with RLS instead of a shared password**
   One password and Upstash Redis were enough for one person. For many accounts, Supabase Auth gives each person a login, and row-level security guards the data in the database rather than only in function code. The old data was moved once into one account with a migration script.
3. **One row per record**
   Each session, check-in, or event is its own row and is written individually. Two devices editing different records never overwrite each other, unlike saving all the data as a single document.
4. **The plan is computed in the browser**
   The server only stores raw data; the plan, warnings, and predictions are recomputed in the browser, so they still work offline. For push notifications the page uploads the next few weeks of the plan so the cron job knows the day's session.
5. **A superadmin only for maintenance**
   The superadmin keeps no training data and cannot open anyone's check-ins, pain, profile, or plan; run history is read through an SQL function that picks its own columns. Superadmins are added only from the SQL Editor.
6. **intervals.icu instead of Strava**
   Huawei watches have no public API, and since June 2026 the Strava API requires a paid subscription. intervals.icu is free, connects to Huawei Health, and provides activities, streams (map, pace, HR), and wellness. Each account's key is kept in a server-only table.

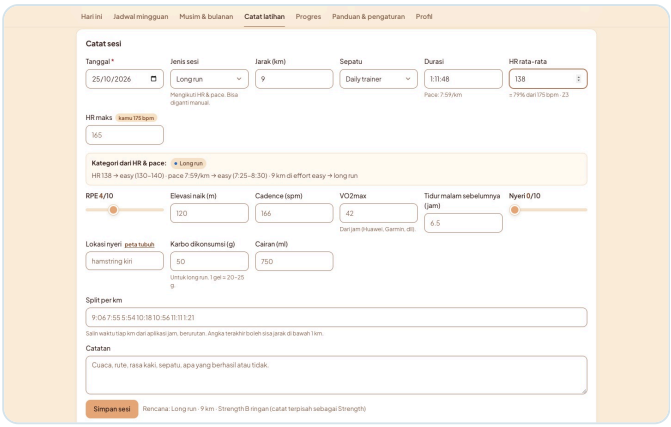
HOW TO USE IT

These steps follow the same order as the tutorial video.



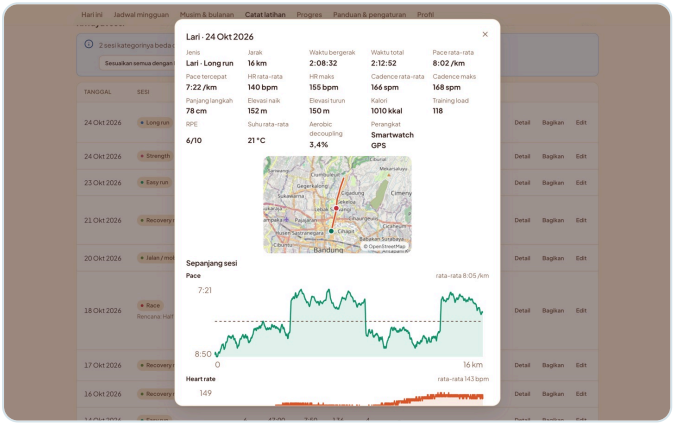
1 Open the Today tab

See today's session for the current phase, load warnings at the top, and the weather and air quality for your next run, then fill in the morning check-in.



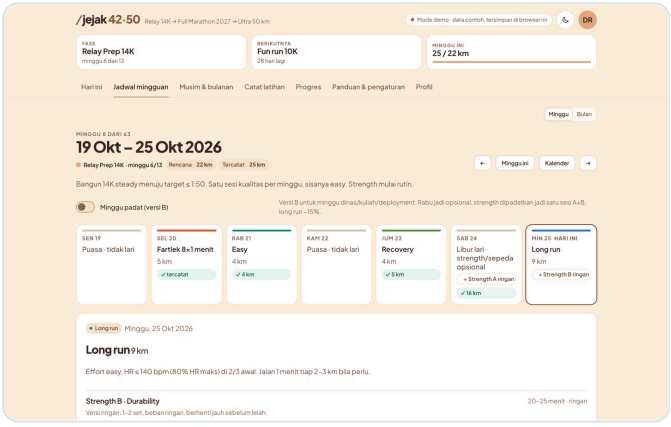
2 Log a session

Press Log this session or open Log training, then enter distance, duration, and HR. Pace and category are calculated for you; press Save session.



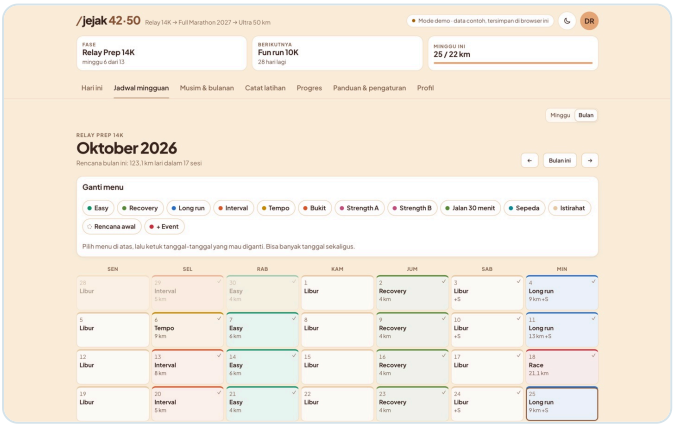
3 Open a watch session's details

Press Detail in the session history. Watch sessions show the map, pace/HR/cadence/elevation charts, HR zones, and per-km splits.



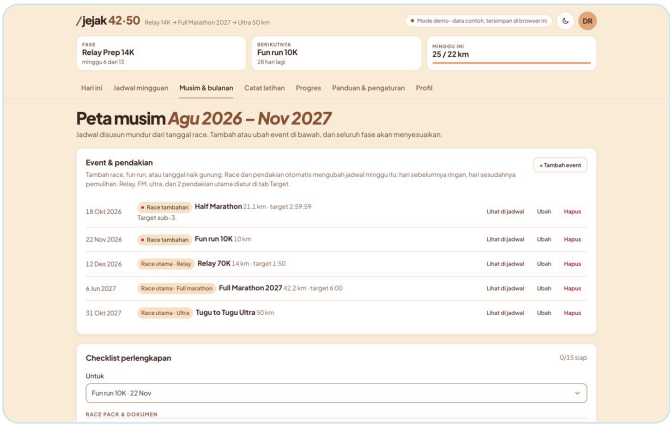
4 Check the weekly plan

The weekly plan tab shows each day's session, the week's km target, and which sessions are done. Tap a day to see or change its menu.



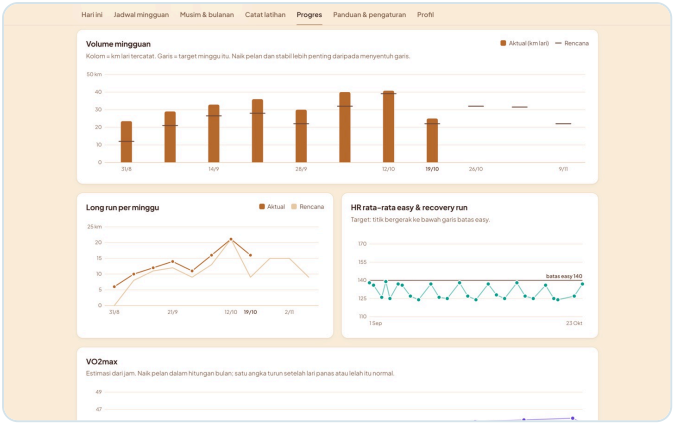
5 Plan a whole month

Choose Month at the top right, pick a menu under Change menu, then tap the dates to change. Tap the menu again to stop.



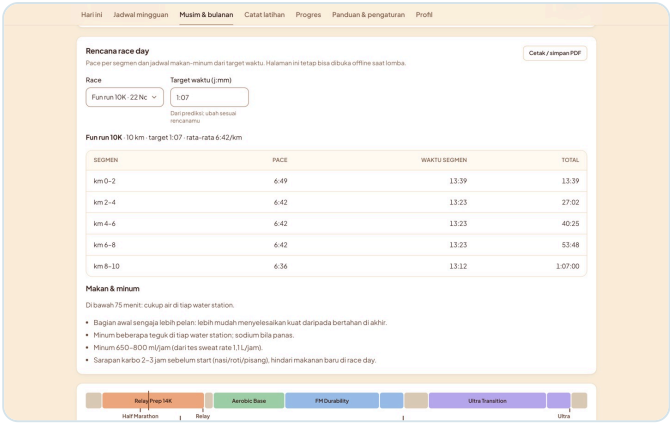
6 See the season & events

In Season & monthly, the season map shows the phases up to the ultra, and the Events & hikes list holds the main and extra races. Add an event to adjust that week's plan.



7 Track progress

The Progress tab has weekly volume, long runs, VO2max, Fitness · Fatigue · Form, achievements, and race predictions.



8 Plan race day

In Season & monthly, pick a race and a target time to get pace per segment and an eating and drinking schedule. The profile keeps records, HR zones, and the pain map; dark mode is the moon button at the top.

LIMITATIONS & NOTES

- The planning and warning rules are personal training heuristics, not medical advice.
- There is no open sign-up: accounts are created by the superadmin. The Gemini quota and VAPID keys are shared by every account.
- The Fitness · Fatigue · Form chart, session detail with a map, and wellness data need intervals.icu sync; manual sessions only show their stats.
- Gemini's free tier allows roughly 20 requests per day across reviews, questions, and imports.
- On the Hobby plan, Vercel Cron runs once a day within the given hour, so notification times are not exact.
- The public demo is a separate build with no API and no accounts: it runs in local mode with sample data that resets daily and has no Target tab (its races are fixed), so watch sync, the AI coach, notifications, and the superadmin page are off there. The sign-in and superadmin screenshots use mock data.

WHAT'S NEXT

- Automated tests for the planning and warning logic in src/lib.
- Letting a coach edit an athlete's plan, not just read it.